

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“Development of Vector DB-Based Chatbot Agent”

Submitted in partial fulfillment for the award of degree(20INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Prakriti Singh - 1DS21AI039

Conducted at



SNTI Tata Steel Ltd.

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Development of Vector DB-Based Chatbot Agent**” carried out by **Ms. PRAKRITI SINGH [1DS21AI039]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Soumyabroto Banerjee
Engineer
Tata Steel

Signature of Guide
Dr. Archudha A
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Prakriti Singh**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Tata Steel Ltd.** This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI039

Place: Bangalore

NAME: Prakriti Singh

OFFER LETTER



Tata Prashikshan - 2025



Dear Prakriti,

We are happy to inform you that your application for Internship has been approved. Your reference number is **VT20250422**.

Please check our web site mentioned below for further details .

Internet Address:-

<https://tslhr.tatasteel.co.in/recruit/VT001.aspx>

Internship details -

Name : PRAKRITI SINGH
Name of Institute : DAYANAND SAGAR COLLEGE OF ENGINEERING
Branch : Computer Science / Information Technology
Start date : 24-JAN-2025
Duration : 25 Weeks
Reporting Time : For Offline gate pass processing Tuesday and Thursday (10:AM) at respective location.
Venue : Project work can be done from respective location either Online or Offline

Regards,

SNTI, TATA STEEL LTD.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

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Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible.

PRAKRITI SINGH [1DS21AI039]

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CHAPTER 1

INTRODUCTION

INTRODUCTION TO VECTOR DB-BASED CHATBOT AGENT

A **Vector Database-Based Chatbot Agent** is an advanced AI system designed to provide accurate, context-aware responses by leveraging the power of **vector embeddings** and **semantic search**. Unlike traditional chatbots that rely on keyword matching or fixed rule-based responses, this system utilizes **vector databases**—such as **FAISS** or **Chroma**—to store and retrieve document embeddings based on their semantic similarity to user queries.

In this project, the chatbot agent is designed to interact with domain-specific documents, particularly those from various departments of **Tata Steel Ltd.**, such as HR and Sales. These documents are first processed and converted into vector embeddings using large language model (LLM) frameworks like **LangChain** and **Google Gemini API**. When a user poses a question, the chatbot searches for the most relevant document chunks in the vector database and generates a response using a **Retrieval-Augmented Generation (RAG)** approach.

This architecture ensures that the chatbot not only understands the context of a query but also responds with information grounded in company-specific documentation. It marks a significant step forward in building **intelligent enterprise assistants** capable of handling complex, department-specific queries with precision and efficiency.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

TATA STEEL LTD.



ABOUT THE COMAPNY

Tata Steel Limited is an Indian multinational steel-making company headquartered in Mumbai, Maharashtra, with its primary operations based in Jamshedpur, Jharkhand. It is a subsidiary of the Tata Group.

Tata Steel is increasingly investing in **Artificial Intelligence** to drive **digital transformation** across its operations. The company uses AI to enhance **operational efficiency, predictive maintenance, supply chain optimization, quality control, and safety management**. AI is also used in **process automation** to reduce downtime and energy consumption, ensuring more sustainable and cost-effective production.

Notably, Tata Steel is exploring the development of **AI-powered agents and chatbots** to assist in internal operations such as **HR automation, sales support, and knowledge management**. These initiatives demonstrate the company's commitment to leveraging AI for smarter decision-making, better customer service, and enhanced workforce productivity.

VISION:

To be the **most respected and efficient steel producer** in the world.

Tata Steel aims to lead the global steel industry not only through volume and profitability but by upholding the highest standards of **ethics, innovation, sustainability, and stakeholder value creation**.

MISSION:

The company's mission is centered around creating lasting value for all its stakeholders. This is achieved through a strong emphasis on operational excellence, fostering innovation to develop advanced solutions, and maintaining an unwavering commitment to sustainability and social responsibility. Tata Steel integrates these principles into every aspect of its operations, reinforcing its role as a forward-thinking and responsible global enterprise.

CORE AREAS OF INNOVATION:

- **Process Innovation:**

One of the primary areas is process innovation, where Tata Steel continuously refines its manufacturing methods to enhance productivity, reduce waste, and minimize environmental impact. This includes the adoption of advanced process control systems and automation.

- **Product Innovation:**

The second area is product innovation, aimed at developing high-performance steel products tailored to evolving market needs, especially in sectors like automotive, construction, and infrastructure. This involves R&D efforts to create lighter, stronger, and more sustainable steel grades.

- **Digital transformation:**

Digital transformation is a third major focus, where the integration of Artificial Intelligence, Machine Learning, IoT, and data analytics is transforming traditional workflows. These technologies are being used for predictive maintenance, smart logistics, and decision intelligence.

- **Sustainability and environmental innovation:**

Tata Steel is committed to reducing its carbon footprint through energy-efficient technologies, recycling initiatives, and exploring green steel production methods.

- **People-centric innovation:**

This includes deploying digital tools to enhance workforce safety, improve training methods, and streamline internal processes such as HR and knowledge management, often through AI-powered systems like chatbots.

CHAPTER 3

SCOPE OF WORK

The scope of work undertaken during the internship at Tata Steel Ltd., under the guidance of the Department of Artificial Intelligence & Machine Learning, Dayananda Sagar College of Engineering, focused on the research, design, and initial development of a Vector Database-Based Chatbot Agent. The core objective was to explore the practical implementation of Retrieval-Augmented Generation (RAG) using large language models and vector databases to build an intelligent, document-aware chatbot tailored for internal enterprise use. The internship provided significant exposure to LLM pipelines, document processing, and semantic search in an enterprise AI context.

1. Familiarization with LangChain and LangFlow (Initial Phase)

The internship began with an extensive review and hands-on exploration of **LangChain** and **LangFlow**, frameworks used for building LLM-driven applications. This phase provided foundational understanding of:

- LangChain modules, chains, and agents for LLM-based task execution
- LangFlow as a visual interface for pipeline design
- Key concepts like prompts, memory, and document loaders
- Integration of Google Gemini API as the LLM backend

By the end of this phase, a working development environment was set up, enabling the design and experimentation of LLM workflows within a modular and scalable architecture.

2. Document Collection and Vector Embedding

Following the setup phase, a major portion of the work focused on the collection and processing of **Tata Steel's departmental documents**, particularly **annual reports** and internal PDFs. Tasks in this stage included:

- Cleaning and segmenting lengthy PDFs into semantically meaningful chunks
- Generating **vector embeddings** using LLM-supported embedding models
- Configuring and populating **FAISS** and **Chroma** vector databases for fast similarity search
- Mapping document content to specific departments for query relevance

This phase was critical in preparing the data layer of the chatbot and ensured that the vector database could accurately respond to queries by retrieving the most relevant information.

3. Development of Sub-Chatbots for Department-Specific Use

A key innovation was the design of **sub-chatbots**, each tailored to specific business units like **HR** and **Sales**. These agents shared a master control logic but accessed department-specific document vectors, enabling targeted responses. This included:

- Designing logic for query routing based on department classification
- Integrating LangChain-based document retrieval pipelines for each sub-chatbot
- Testing query accuracy and evaluating semantic search performance

This approach enabled modular, scalable architecture suitable for large enterprises with diverse information silos.

4. Integration of Google Gemini API for Response Generation

Once the vector database and routing logic were functional, the project moved to integrating the **Gemini API** for dynamic query response generation. This included:

- Configuring Gemini as the LLM endpoint within LangChain
- Constructing custom prompts and templates for context-aware replies
- Testing LLM behavior with both generic and department-specific queries
- Iteratively tuning prompts to improve output relevance and readability

This integration allowed the chatbot to combine retrieved information with generative capabilities, forming the backbone of the Retrieval-Augmented Generation system.

The internship enabled immersive exposure to modern AI technologies and best practices in real-world enterprise contexts. By engaging in tasks spanning data processing, vector embedding, LLM pipeline design, and chatbot development, the project offered a comprehensive experience in building intelligent agents for internal business applications. The work lays a strong foundation for future enhancements, including UI development and full-scale deployment within Tata Steel's digital ecosystem.

CHAPTER 4

LEARNING OBJECTIVES

1. Understanding the Retrieval-Augmented Generation (RAG) Architecture:

A primary learning objective of the internship was to gain a solid understanding of the **Retrieval-Augmented Generation (RAG)** architecture, a modern AI approach that enhances large language model responses by grounding them in external knowledge sources. This involved exploring how vector databases work in tandem with LLMs to retrieve relevant document fragments and generate accurate, context-aware responses. By implementing this system in the context of Tata Steel's internal documentation, the internship offered hands-on exposure to one of the most advanced and practical applications of generative AI in enterprise environments.

2. Gaining Practical Knowledge of Vector Databases and Semantic Search:

Another key objective was to develop practical skills in working with **vector databases** such as **FAISS** and **Chroma**, which are essential for enabling **semantic search** capabilities. The internship involved converting departmental documents into vector embeddings, indexing them in a database, and retrieving them based on similarity to user queries. Through this, concepts such as chunking strategies, cosine similarity, and document retrieval pipelines were understood in-depth, along with the practical challenges of dealing with real-world, domain-specific textual data.

3. Hands-On Experience with LangChain, LangFlow, and Google Gemini API:

A significant goal of the internship was to learn and apply **LangChain**, a framework for building LLM-powered applications, and **LangFlow**, its visual interface counterpart. Tasks involved creating agent chains, designing prompts, handling memory, and constructing custom logic flows using LangFlow's interface. Furthermore, integration with the **Google Gemini API** provided insight into real-time LLM interaction, fine-tuning of prompts, and interpreting outputs for different types of queries. This experience was instrumental in building a foundation for future work in LLM-based software solutions.

4. Learning Document Processing and Embedding Workflows:

The internship also aimed to develop proficiency in **document preprocessing**, including cleaning, parsing, and segmenting large, complex documents such as annual reports. These documents were embedded into vector space using modern embedding models to facilitate efficient semantic search. This process provided a practical understanding of how unstructured data can be structured and indexed for high-quality machine understanding and retrieval—an essential skill in modern enterprise AI applications.

5. Exploring Enterprise AI Applications and Real-World Challenges:

Finally, a key objective was to explore how artificial intelligence is being applied in real-world enterprise settings, particularly in large-scale organizations like **Tata Steel Ltd.** The project revealed challenges such as tool compatibility, deployment scalability, environment setup, and document variability. Overcoming these hurdles taught the importance of **collaboration**, **iteration**, and **mentor guidance** in the development process. The experience bridged the gap between academic AI theory and industry-grade AI implementation, highlighting the potential and complexity of building intelligent systems in structured corporate ecosystems.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Understanding and Implementing LangChain Modules

Challenges:

One of the early challenges was understanding how to effectively use **LangChain**, a framework designed for building applications powered by large language models. As a beginner to the platform, it was initially difficult to grasp the structure and purpose of its components, such as **chains**, **agents**, **memory modules**, and **document loaders**. Misunderstanding the flow often led to improper configuration of pipelines, resulting in non-functional chains or inaccurate outputs during testing.

Solution:

To overcome this, official LangChain documentation, community tutorials, and example notebooks were studied in depth. A bottom-up learning approach was adopted—starting from simple chains and gradually moving to complex multi-component workflows. Code snippets were isolated and tested individually to understand their functionality. LangFlow, the visual version of LangChain, was also leveraged to visualize logic flow, making debugging easier. With regular feedback from mentors, a working knowledge of LangChain’s modular architecture was successfully developed, laying a strong foundation for the rest of the project.

2. Selecting and Configuring the Right Vector Database

Challenges:

Choosing between **FAISS** and **Chroma** as the vector database for embedding and retrieval presented both technical and performance-related difficulties. Each had different strengths, and initial experiments often led to poor retrieval quality or slow query responses. Understanding the trade-offs—such as indexing speed, storage requirements, and API limitations—proved to be time-consuming.

Solution:

To resolve this, both vector stores were tested with a common set of document embeddings. Benchmarking was done on metrics such as retrieval accuracy, query response time, and ease of integration with LangChain. Chroma was ultimately chosen for its developer-friendly setup and dynamic collection management. Custom chunking strategies and embedding models were applied to enhance semantic matching. By iterating over indexing configurations and vector similarity metrics, a robust and efficient retrieval system was achieved.

3. Processing Long and Complex PDF Documents

Challenges:

The source documents—particularly Tata Steel’s annual reports—were long, complex, and unstructured, making them challenging to process effectively for chatbot integration. Issues included inconsistent formatting, lengthy paragraphs, and embedded tables or charts that disrupted chunking logic. This often led to poor-quality vector embeddings and irrelevant search results.

Solution:

To tackle this, a multi-step preprocessing pipeline was implemented. Documents were first converted to plain text using reliable PDF parsers, followed by custom scripts to clean and normalize the data. A chunking strategy based on semantic boundaries (rather than fixed lengths) was developed using sentence tokenizers. Sample documents were used to test embedding quality and adjust the logic accordingly. This preprocessing overhaul significantly improved the relevance and coherence of the chatbot’s responses.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Python (with libraries: Pandas, NumPy, PyPDF2)
- LangChain (LLM application framework)
- LangFlow (visual interface for LangChain pipelines)
- Google Gemini API
- FAISS / Chroma (Vector Databases)
- Jupyter Notebook (for experimentation and development)
- Google AI Studio (for Gemini API configuration and testing)
- Google Docs / Sheets (for documentation and collaboration)
- Git & GitHub (for version control and code sharing)
- Slack / Microsoft Teams / Email (for mentor communication)

Hardware Requirements:

- Processor: Intel i5 or more
- RAM: 8 GB minimum
- Display: 23 LED Display
- 64bit operating systems
- LAN Connection Port: Required

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Environment Setup & LangChain Familiarization (Initial Phase):

During the early phase of the internship, significant effort was dedicated to setting up the **development environment** and gaining hands-on familiarity with the **LangChain** and **LangFlow** frameworks. Initial tasks involved configuring the necessary Python libraries, establishing project structures in Jupyter Notebooks, and integrating LangChain with **Google Gemini API**. This foundational work laid the groundwork for future development and allowed a deeper understanding of how chains, agents, and memory components operate in a modular LLM framework. Challenges with version conflicts and missing dependencies were resolved through iterative debugging and mentorship support. This phase established a strong technical base for the advanced components of the chatbot system.

2. Document Preprocessing and Vector Database Integration:

One of the core contributions was the development of a robust **document preprocessing pipeline** to handle long and unstructured enterprise PDFs—specifically **Tata Steel’s annual reports** and internal documents. The documents were converted into clean, semantically coherent chunks using custom tokenization scripts, and these chunks were embedded into a **vector space** using state-of-the-art embedding models. The embedded content was then indexed in **Chroma** and **FAISS** vector databases to enable efficient semantic retrieval. A key milestone was successfully querying the vector store and retrieving relevant document sections based on natural language input, demonstrating end-to-end data flow from static text to intelligent response.

3. Sub-Chatbot Architecture for Department-Specific Query Handling:

A significant architectural contribution was the design and implementation of **sub-chatbots** tailored for different departments such as **HR** and **Sales**, under a unified master agent. Each sub-chatbot was configured to reference only its department’s relevant documents, improving the specificity and relevance of the responses. Custom logic was developed to **route queries** to the appropriate sub-agent based on keywords and context. This modular design introduced scalability and maintainability to the chatbot system, and set the foundation for enterprise-wide deployment where multiple departments can independently leverage LLM-powered query handling.

4. Integration of Gemini API with LangChain Pipelines:

A key technical achievement was the **successful integration of Google's Gemini API** as the LLM backend within LangChain pipelines. This involved API key authentication, creating custom prompt templates, and embedding the language model into the agent workflow. The LLM was configured to generate dynamic, context-aware responses based on the information retrieved from the vector database. Several iterations were conducted to refine the prompt structure for improved clarity and relevance in responses. This integration marked the transition from static data retrieval to a fully functional **Retrieval-Augmented Generation (RAG)** system capable of delivering intelligent, grounded answers in real time.

5. Workflow Design Using LangFlow:

Visual pipeline development using **LangFlow** played a critical role in validating and debugging the logic of the chatbot system. By designing chains visually, the project benefited from a clearer understanding of the data flow between components such as loaders, retrievers, prompts, and LLMs. LangFlow was also used to demonstrate chatbot logic to mentors and stakeholders, bridging the gap between technical complexity and stakeholder communication. Visual workflows ensured quick iteration and modular testing, ultimately accelerating development progress and improving design accuracy.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at **Tata Steel Ltd.**, conducted under the guidance of the **Department of Artificial Intelligence & Machine Learning at Dayananda Sagar College of Engineering**, has been an intellectually rewarding and transformative experience. Working on the **Vector Database-Based Chatbot Agent** provided hands-on exposure to modern AI tools and methodologies, particularly in the areas of **LLM integration**, **semantic search**, and **document-aware chatbot design** for enterprise use cases.

One of the most impactful aspects of the internship was the opportunity to work at the intersection of **natural language processing** and **information retrieval**—an emerging area with real-world relevance. From understanding how to preprocess and embed enterprise documents to configuring vector databases and building LangChain pipelines, the project enabled a comprehensive understanding of the full Retrieval-Augmented Generation (RAG) pipeline. This real-world exposure bridged the gap between academic theory and practical application, reinforcing concepts through iterative problem-solving.

The freedom to explore advanced tools such as **LangChain**, **LangFlow**, and the **Google Gemini API** fostered an experimental mindset and enhanced my ability to work independently with modern AI development frameworks. Designing sub-chatbots for different departments added a layer of complexity and taught me the importance of **modular architecture**, **context management**, and **information scope isolation** in building scalable AI solutions for large organizations.

Collaboration with mentors and the emphasis on documentation and code clarity also highlighted the importance of **professional communication** and **team coordination**—skills that are just as crucial as technical proficiency in enterprise environments. Regular feedback sessions and troubleshooting discussions helped refine both my technical approach and my ability to convey complex ideas effectively.

The challenges faced—ranging from API configuration and document chunking to vector DB selection and pipeline debugging—served as powerful learning opportunities. Each issue required a thoughtful, layered approach, reinforcing the importance of structured testing, patience, and adaptability when dealing with enterprise-grade AI systems.

CHAPTER 8

RECOMMENDATIONS

1. **Implement User Interface for Chatbot Interaction:** To improve accessibility and usability, it is recommended to develop a simple yet intuitive user interface for the chatbot. A web-based dashboard or lightweight application would allow end users—such as HR personnel or sales executives—to interact with the system without needing to run backend scripts or access code-level tools.
2. **Department-Level Document Expansion:** While the current system supports basic department-specific sub-chatbots, future improvements should focus on incorporating a wider variety of internal documents, such as policies, training manuals, and FAQs. Expanding the dataset will increase the chatbot’s knowledge base and improve its ability to answer diverse and complex queries accurately.
3. **Integrate Logging and Analytics for Query Monitoring:** Adding a logging mechanism to track user queries and chatbot responses can offer valuable insights into usage patterns, frequent questions, and performance gaps. This data can support iterative improvements and allow for better debugging and optimization of the chatbot pipelines.
4. **Enhance Prompt Engineering for Improved Response Accuracy::** While the chatbot currently uses basic prompt templates for interacting with the Gemini API, more sophisticated prompt engineering techniques—such as dynamic prompt injection and multi-turn memory handling—could enhance response relevance and maintain conversational context over longer interactions.
5. **Introduce Access Management and Content Filtering:** As the chatbot scales to handle sensitive organizational documents, it is advisable to implement access control mechanisms and content filtering. Ensuring that each user can only retrieve department-relevant and permission-appropriate data will help maintain information privacy and security within the enterprise environment.

CHAPTER 9

CONCLUSION

The internship at Tata Steel Ltd., in collaboration with the Department of Artificial Intelligence & Machine Learning at Dayananda Sagar College of Engineering, has been a deeply enriching experience that provided meaningful exposure to the practical implementation of cutting-edge AI technologies in an enterprise setting. Working on the Vector DB-Based Chatbot Agent enabled firsthand learning in the fields of natural language processing, vector databases, and LLM-based query systems, all within the framework of a real-world corporate application.

Contributing to every phase of the project—from setting up development environments and learning LangChain to designing department-specific chatbot logic, processing large-scale enterprise documents, and integrating the Gemini API—offered a well-rounded technical and strategic experience. Each task demanded not only technical competence but also adaptability, problem-solving, and the ability to collaborate across different tools and platforms.

The internship offered deep insight into how large enterprises like Tata Steel are leveraging AI to streamline internal workflows, improve access to knowledge, and enhance decision-making efficiency. The challenges encountered—whether related to document chunking, pipeline debugging, or tool configuration—provided invaluable lessons about working with complex, unstructured data and integrating AI into existing organizational ecosystems.

In addition to the technical gains, the experience cultivated important soft skills such as clear communication, time management, and the ability to learn independently. Mentor guidance and collaborative discussions were instrumental in bridging the gap between theoretical knowledge and practical application, especially in implementing real-time, retrieval-augmented solutions.

Overall, this internship has not only strengthened core competencies in AI and software development but also deepened a genuine interest in building intelligent, scalable solutions for enterprise use. It has reaffirmed a commitment to contributing to the evolving landscape of applied AI, where innovation meets real-world utility in meaningful and impactful ways.

CHAPTER 10

REFERENCES

1. LangChain. (2024). *LangChain Documentation*. Retrieved from <https://docs.langchain.com>
2. Chroma. (2024). *Chroma: Open-Source Embedding Database*. Retrieved from <https://www.trychroma.com>
3. Google. (2024). *Gemini API Overview*. Retrieved from <https://ai.google.dev/gemini>
4. FAISS. (2023). *Facebook AI Similarity Search (FAISS)*. Retrieved from <https://github.com/facebookresearch/faiss>
5. Tanwani, A., & Kumar, P. (2022). *A Survey on Retrieval-Augmented Generation in Natural Language Processing*. *Journal of Machine Learning Research*, 23(178), 1–25.SP4 Ameya Innovation Labs Pvt. Ltd. (2024). *Official Website*. Retrieved from <https://www.sp4ameya.com>
6. Tata Steel Ltd. (2024). *Annual Reports and Corporate Information*. Retrieved from <https://www.tatasteel.com>